Microsoft Elevate

CAPSTONE PROJECT

INDIA CRUDE OIL INTELLIGENCE DASHBOARD

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

Currently, India imports approximately 89.2% of its crude oil requirements, making it one of the most import-dependent major economies in the world. It is important to accurately track and understand this dependency to ensure energy security and informed policy-making. India's crude oil imports have grown from 189 MMT in FY2014 to approximately 240 MMT in FY2025, with significant geopolitical shifts reshaping supply sources. Eventually, monitoring the balance between refining capacity, stock levels, and daily consumption patterns becomes a major concern for energy planners. The crucial part is the comprehensive analysis and visualization of these crude oil import trends, supply diversification, and consumption patterns over the past decade to support strategic energy decision-making.

PROPOSED SOLUTION:

- The proposed system aims to address the challenge of analysing and visualizing India's crude oil import dependency over the past decade. This involves leveraging data analytics and visualization techniques to identify trends and patterns accurately. The solution will consist of the following components:
- **Data Collection:**
  - Gather historical data on India's crude oil imports from FY2014 to FY2025, including yearly import volumes, country-wise supply shares, and refinery data.
  - Utilize official government data sources such as PPAC (ppac.gov.in), Ministry of Petroleum and Natural Gas (MoPNG), EIA, and Indian Oil Corporation to ensure data accuracy.
- **Data Preprocessing:**
  - Clean and preprocess the collected government data using Power Query to handle missing values, standardize fiscal year formats, and resolve inconsistencies across multiple sources.
  - Feature engineering to extract relevant calculated fields such as year-over-year growth rates, import dependency ratios, and daily consumption metrics from the raw data.

- **Machine Learning Algorithm (Data Modelling & Analytics):**
  - Implement data modelling and analytical features within Power BI to analyse trends based on historical patterns of crude oil imports, refining capacity, and consumption.
  - Consider incorporating other factors like geopolitical shifts in supply sources and refinery expansion to improve analytical accuracy.
- **Deployment**:
  - Develop a user-friendly Power BI dashboard interface that provides comprehensive insights into India's crude oil import trends, supply sources, and consumption patterns.
  - Deploy the solution as a multi-page Power BI report with interactive filters, considering factors like visual clarity, user accessibility, and cross-filtering between pages.
- **Evaluation**:
  - Assess the dashboard's performance using appropriate metrics to effectively visualize key data points, such as the 89.2% import dependency ratio and 240 MMT import volume.
  - Fine-tune the visual model based on feedback and continuous monitoring of the analytical insights.

SYSTEM APPROACH:

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the India Crude Oil Intelligence Dashboard.

- **System requirements:**
- **Software:** Microsoft Power BI Desktop for data modelling and visualization, Microsoft Excel for initial dataset inspection, and a modern web browser for viewing published reports.
- **Hardware:** A standard computer or laptop capable of smoothly running Power BI Desktop (e.g., Windows 10/11 OS, minimum 4GB RAM, 8GB+ recommended for efficiently processing the multi-table data model).
- **Data Source:** Official government datasets from PPAC (ppac.gov.in), Ministry of Petroleum and Natural Gas, EIA, and Indian Oil Corporation containing crude oil import volumes, country-wise shares, refinery data, and consumption statistics.

SYSTEM APPROACH:

Library required to build the model:

- **Power Query (M Language):** The essential data preparation technology required to clean, transform, and load the raw government datasets into Power BI. Used for merging multiple source files, standardizing fiscal year formats, and creating calculated columns.
- **DAX (Data Analysis Expressions):** The primary formula language required to build custom calculated columns and measures, enabling the calculation of specific metrics like import dependency ratio (89.2%), year-over-year growth rates, and daily consumption averages.
- **Power BI Standard Visuals Library:** The built-in visualization components necessary to render the dashboard charts, including clustered bar charts for import volumes, line charts for trends, stacked bar charts for country-wise shares, and card visuals for KPI metrics. A custom JSON dark theme was applied for consistent styling.

ALGORITHM & DEPLOYMENT:

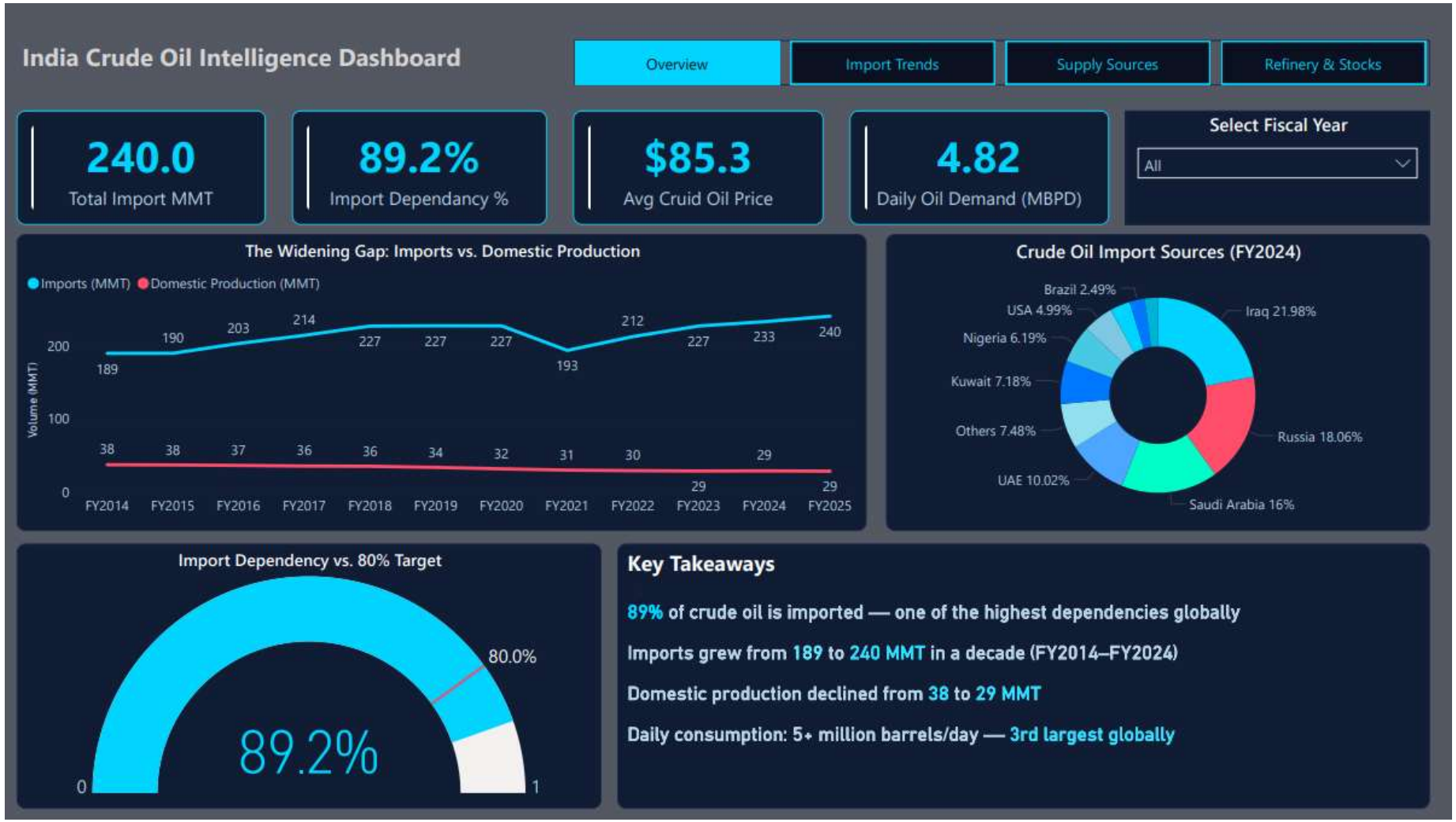
This section describes the algorithms and analytical techniques used in the dashboard, along with the deployment approach for delivering actionable insights.

- **Algorithm Selection:**
  - The dashboard employs DAX-based analytical measures rather than traditional ML algorithms. Key calculations include Year-over-Year growth rates, import dependency ratios, moving averages for trend smoothing, and percentage share calculations for country-wise supplier analysis.
- **Data Input:**
  - Six structured CSV datasets sourced from PPAC covering crude oil imports (country-wise volumes FY2014-FY2025), domestic production and consumption statistics, refinery throughput and capacity data, and petroleum product stock levels. Data was cleaned and transformed using Power Query M language.

- **Training Process:**
  - In a Power BI environment, the "training" phase translates to data modelling - establishing relationships across the 6 tables using a star schema with an FY_Lookup dimension table linking fiscal years.
- **Prediction Process:**
  - The analytical model identifies trends such as Russia's rise to the 2nd largest supplier post-2022, the consistent dominance of Iraq and Saudi Arabia, and the growth from 189 MMT to 240 MMT over the decade.

RESULT:

The following dashboard pages present the key findings of the India Crude Oil Intelligence Dashboard built in Power BI.

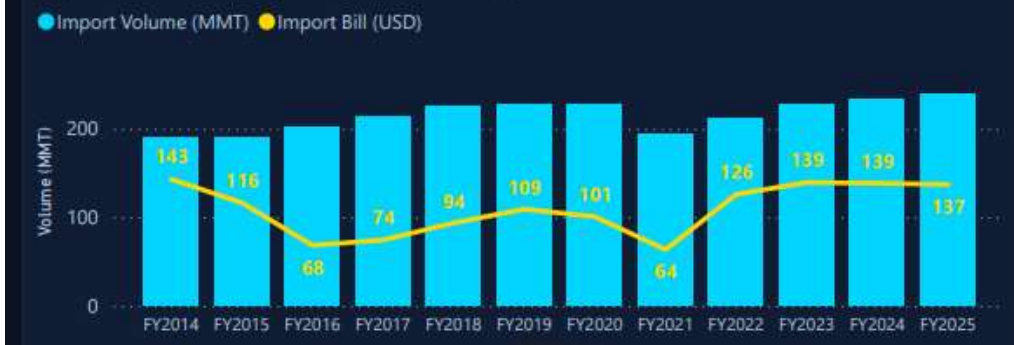


India Crude Oil Intelligence Dashboard

Overview | Import Trends | Supply Sources | Refinery & Stocks

India's Crude Oil Imports: A Decade of Growth (FY2014–FY2025)

Import Volume vs. Import Bill: The Price Impact



Import Dependency Rising: 83% to 89% Over a Decade

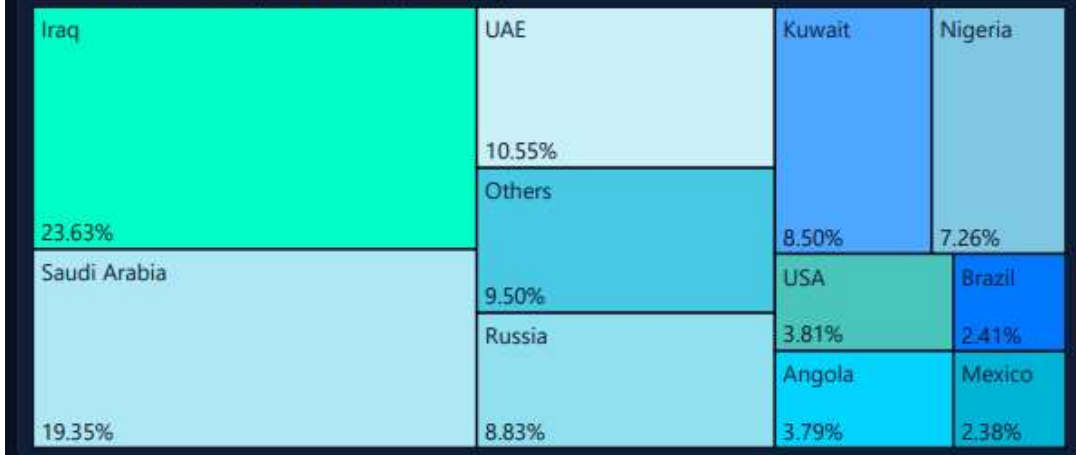
Key Events in India's Crude Oil Timeline

- FY2021: COVID-19 demand crash — imports dropped to 196 MMT
- FY2023: Russia-Ukraine war — Russia becomes #2 supplier, cheap Urals crude floods in
- FY2025: Record imports exceed 240 MMT as demand surges to 5.6 MBPD

India Crude Oil Intelligence Dashboard

Overview | Import Trends | Supply Sources | Refinery & Stocks

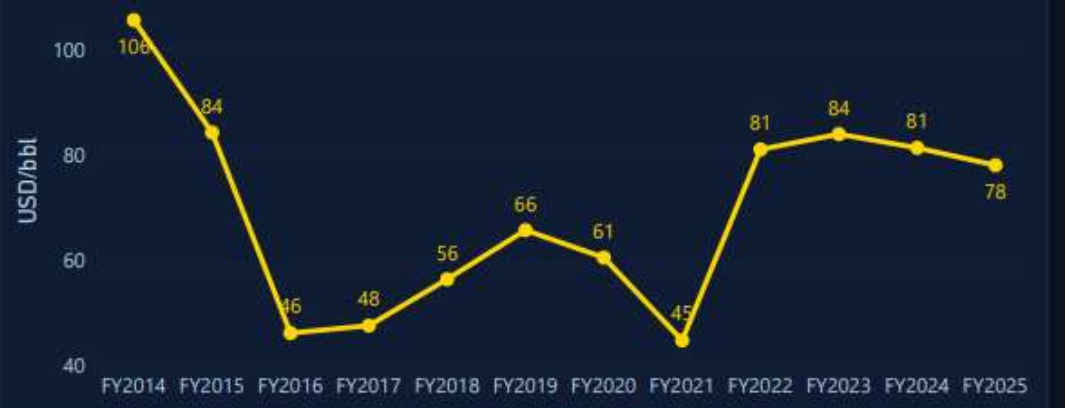
India's Crude Oil Import Share by Country



Russia's Dramatic Rise as India's #2 Supplier

Regional Import Split

Indian Basket Crude Price (USD/bbl)



India Crude Oil Intelligence Dashboard
Overview
Import Trends
Supply Sources
Refinery & Stocks
LPG Stock %
0
78.01
80
100
Petrol Stock %
0
86.19
80
100
Diesel Stock %
0
89.81
80
100
Crude Stock %
0
65.35
80
100
ATF Stock %
0
60.64
80
100
Average Refinery Utilization by Company (%)
100
95
90
85
80
Optimal 95%
98.2
96.2
96.2
96.2
96.1
96.1
96.1
96.1
ONGC
HPCL
IOCL
BPCL
CPCL
MRPL
NRL
RIL
258.12
Total Refining Capacity
23 refineries across India – Source: PPAC
Total Supply vs. Imports (MMT)
Supply - Refinery Output (MMT)
Imports (MMT)
300
250
189
227
190
227
203
240
214
250
227
262
227
261
227
259
193
224
212
242
227
256
233
262
240
269
FY2014
FY2015
FY2016
FY2017
FY2018
FY2019
FY2020
FY2021
FY2022
FY2023
FY2024
FY2025

CONCLUSION:

The Power BI dashboard effectively analyses India's crude oil import dependency over the past decade, providing clear insights into critical energy trends. Key findings reveal that India's import dependency stands at approximately 89.2%, with total crude oil imports growing from 189 MMT in FY2014 to approximately 240 MMT in FY2025. The visual data clearly establishes that Iraq, Saudi Arabia, and Russia are the top three suppliers, with Russia's share rising dramatically post-2022 due to geopolitical shifts. A significant challenge highlighted during this analysis is the concentration risk in supply sources and the gap between domestic refining capacity (258 MMTPA) and growing consumption (5+ MBPD daily). Potential improvements for this analytical solution could include integrating real-time commodity pricing data and forecasting models to predict future import costs and supply disruptions.

Ultimately, this project emphasizes the crucial importance of accurate analytics andvisualization of crude oil import data for India's energy security planning. Understandingthese specific import trends, supply diversification patterns, and consumption metricsis essential for policymakers and energy planners to ensure a stable, secure, and cost-effective crude oil supply chain for the nation.

FUTURE SCOPE:

Discussing potential enhancements and expansions for the system is vital for the next phase of this analytics project. Future iterations could include incorporating additional data sources to expand beyond the current government datasets, perhaps integrating real-time crude oil spot prices and futures data from commodity exchanges. Expanding the system to cover downstream petroleum products (petrol, diesel, LPG) would provide a much deeper understanding of how India's refining output meets domestic demand across different product categories. Additionally, tracking geopolitical risk indicators over time could help in forecasting potential supply disruptions from key source nations. Furthermore, future development should consider the integration of emerging technologies, such as advanced machine learning techniques, to build predictive models for import volume forecasting. These models could dynamically forecast shifts in supply patterns, predicting how quickly alternative suppliers might emerge to reduce concentration risk. Incorporating Power BI Service for cloud deployment with automated data refresh could also be explored to provide stakeholders with always-current intelligence for strategic energy planning.

REFERENCES:

Data Source: Petroleum Planning and Analysis Cell (PPAC), Ministry of Petroleum and Natural Gas, Government of India (ppac.gov.in). This was the primary data source for crude oil import volumes, refinery data, and consumption statistics. Additional data sourced from Energy Information Administration (EIA) and Indian Oil Corporation annual reports.

Primary Source — PPAC (Petroleum Planning and Analysis Cell)

•PPAC Homepage: https://ppac.gov.in

•Import/Export of Crude Oil & Petroleum Products: https://ppac.gov.in/import-export — Country-wise crude oil import volumes, export data

•Consumption of Petroleum Products: https://ppac.gov.in/consumption/products-wise — Product-wise domestic consumption data

•Installed Refinery Capacity: https://ppac.gov.in/infrastructure/installed-refinery-capacity — Refinery-wise capacity data (the April 2025 Excel file was downloaded from here)

•Crude Processing: https://ppac.gov.in/production/crude-processing — Refinery throughput data

•Location of Refineries: https://ppac.gov.in/infrastructure/location-of-refineries

GitHub :

- https://github.com/Viswanathan49/india-crude-oil-intelligence-dashboard

Thank You